

PAHDD: A Port-Based Adaptive High-Speed Traffic DDoS Real-Time Detection System

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Abstract—With the increase in Internet-connected devices, Distributed Denial of Service (DDoS) attacks pose a growing threat to Internet security. Experts in the field have conducted extensive research. However, most existing methods require extracting IP addresses for DDoS detection. This strategy is not efficient when protected network segments utilize NAT technology and the destination IP information is not accessible due to strict confidentiality constraints. To address this issue, we propose a port-based adaptive high-speed traffic DDoS real-time detection (PAHDD) system. The PAHDD system adaptively locks high-risk attacked ports, isolates services based on their susceptibility to high-rate and low-rate DDoS attacks, and performs hierarchical detection of these attacks. We implemented and trained the PAHDD system using the training dataset from the Aitrans2024 competition and evaluated it on the test dataset, achieving an accuracy of 99.87%. PAHDD utilizes port-based detection instead of relying on the IP information of protected network segments, providing a unique method for detecting DDoS attacks in high-privacy network environments.

Index Terms—DDoS; Port-based; Detection; NAT

I. INTRODUCTION

As the Internet evolved, network traffic has increased on an indicative scale. While the Internet provides significant convenience to people, it also makes cybersecurity increasingly important and challenging. In a variety of cyber-malware attacks, Distributed Denial of Service (DDoS) attacks pose a huge threat to the network and are hard to defend.

Denial of Service (DoS) is a malicious network attack method where a target system is bombarded with a large number of invalid requests, exhausting or paralyzing its resources, and consequently preventing legitimate users from accessing the system. DDoS attacks involve multiple computers working together to form an attack platform, simultaneously launching DoS attacks against one or more targets, thus exponentially increasing the potency of a DoS attack [1]. Since its emergence, DDoS has been favored by attackers, evolving into hundreds of different attack methods. Experts in the field have proposed numerous detection methods for various types of DDoS attacks. However, the vast majority of these methods heavily rely on the source/destination IP addresses of the traffic, which complicates the detection of DDoS traffic in high-privacy network scenarios, including those utilizing Network Address Translation (NAT) technology [2].

In network scenarios using NAT technology, the inability to obtain destination IP information and the anonymization of traffic data significantly complicate real-time detection. This

limitation results in suboptimal performance of existing DDoS detection methods in such networks. Companies like Alibaba Cloud have proposed solutions with basic DDoS protection capabilities for systems using NAT technology, including threshold setting and traffic scrubbing. However, due to bandwidth limitations and reliance on packet characteristics, these systems lack adaptive capabilities and are ineffective against large-scale traffic attacks. Therefore, new detection approaches are needed for these specific network environments [3].

The thesis of this paper unfolds based on the following motivations:

Motivation 1: Existing algorithms fail to meet the requirements of scenarios demanding high confidentiality of IP information. Current detection methods can be classified into two categories: those based on IP addresses and those not based on IP addresses. The former relies on information derived from destination IP addresses, making it difficult to adapt to network scenarios where destination IP information is not provided. The latter employs machine learning or wavelet detection methods based on other packet characteristics but cannot effectively detect hybrid attacks. Therefore, high-privacy networks require a detection approach that does not depend on destination IP information and can effectively defend against hybrid attacks.

Motivation 2: Existing algorithms lack the perspective of isolating services. In real-world attack scenarios, a specific port on an attacked server might be targeted while other ports continue to operate normally. In such cases, subdividing the different services provided by the same server ensures that the attacked server does not shut down other well-functioning services. To achieve this, it is necessary to classify attack packets based on ports and thereby isolate different services.

To address the aforementioned issues, we propose a port-based adaptive high-speed traffic DDoS real-time detection (PAHDD) system. The contributions of this paper are as follows:

- We designed the PAHDD system. The PAHDD system classifies traffic into three different states based on the tuple $\langle src_port, dst_port \rangle$, achieving hierarchical detection of high-rate DDoS (HDDoS) and low-rate DDoS (LDDoS) attacks and effective isolation of services with different risk types. In particular, the LDDoS detection module employs a random forest algorithm to accurately identify traffic that closely resembles background traffic.

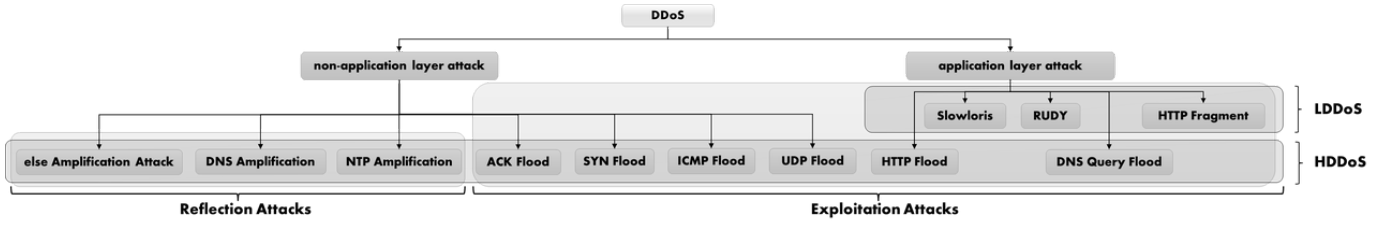


Fig. 1. Classification of DDoS Attack methods.

- We innovatively detect DDoS attacks from the perspective of ports and design an adaptive risk port locking algorithm. By monitoring the traffic of source and destination ports, the system adaptively categorizes ports into HDDoS risk ports, LDDoS risk ports, and benign ports. Incoming traffic is initially directed to different branches, eliminating the need for all traffic to traverse every module in the system, significantly reducing detection time and average latency.
- We implemented the PAHDD system, trained it using the Aitrans2024 competition training dataset [16], and evaluated it on the test dataset. Experimental results show that our system achieved an *accuracy* of 99.87%, a *precision* of 93.93% for predicting positive classes, and an *F1 – score* of 89.01%.

The rest of this paper is organized as follows: Section 2 reviews related work, Section 3 introduces the design of the method, Section 4 details the system implementation and evaluation, and Section 5 concludes the paper.

II. RELATED WORK

A. Classification of DDoS Attacks

There are many ways to classify DDoS attacks. As illustrated in Fig. 1, DDoS attacks can be categorized by network protocol stack layers into application layer attacks and non-application layer attacks. Another classification is based on the involvement of third parties, distinguishing between resource exhaustion attacks and reflection amplification attacks. Resource exhaustion attacks aim to incapacitate the target server by depleting its resources, while reflection amplification attacks use third-party servers to amplify the attack traffic, thereby masking the attacker’s identity. Additionally, DDoS attacks can be classified by packet rate into high-rate and low-rate attacks [4].

B. DDoS Attack Detection Methods

Currently, researchers have proposed numerous methods for detecting DDoS attacks. While a few algorithms do not use source/destination IP as key features, the majority still heavily rely on IP information. These methods can be broadly categorized into five types: machine learning-based methods, probabilistic methods, entropy-based methods, signal processing-based methods, and graph-based methods.

1) *IP Based*: Experts in this field have provided numerous effective methods for detecting various types of DDoS attacks.

a) *Machine Learning Based Approach*: J. J. Villalobos et al. proposed an unsupervised method based on memory-distributed graphs and stream data analysis. The algorithm combines traditional statistical methods with the K-means clustering algorithm to create a DDoS detection system based on packet features such as IP addresses [5].

b) *Statistically Based Approach*: Jisa & Ciza developed a statistical algorithm that uses IP-related features within a time window, employing traffic characteristics and dynamic thresholds for detection [6]. Ke Li et al. introduced a hybrid metric method by combining the Total Variation Metric and the Bhattacharyya Coefficient, which measures the differences and similarities between two discrete probability distributions [7].

c) *Information Entropy Based Approach*: Monowar H. Bhuyan et al. evaluated the effectiveness of Hartley entropy, Shannon entropy, Renyi entropy, and generalized entropy in detecting low-rate DDoS attacks by comparing these measures with set thresholds [8]. Monika Sachdeva et al. proposed a detection method using traffic cluster entropy and source address entropy, comparing them with thresholds calibrated through ROC curves [9].

d) *Signal Processing Based Approach*: S. Toklu and M. Şimşek presented a dual-layer detection method to simultaneously identify and filter hybrid high-rate and low-rate DDoS attacks. The first layer uses a Discrete Average Filter (DAF) to detect high-rate DDoS attacks, while the second layer employs Discrete Fourier Transform (DFT) to detect low-rate DDoS attacks [10].

e) *Graph Structure Based Approach*: Xinqian Liu et al. designed a multi-dimensional sketch structure using IP addresses to aggregate and compress network traffic for the detection of low-rate DDoS attacks (LDDM). Additionally, they proposed an improved behavioral difference measurement method based on the Daubechies 4 wavelet transform, and utilized an enhanced weighted exponential moving average method to construct dynamic thresholds [11]. Yingjie Zhou et al. developed a detection system that constructs temporal graphs using IP entropy values to accurately describe the relationships among multiple time series for traffic anomaly detection [12].

Although these methods cannot be effective in high-privacy networks due to their excessive reliance on IP address information, they still provide valuable insights for our subsequent research and design.

2) *Non-IP Based*: A minority of non-IP-based detection methods have successfully identified DDoS traffic using techniques such as machine learning, deep learning, wavelet analysis, and power spectral density analysis. Xiaoyong Yuan et al. employed Long Short-Term Memory (LSTM) algorithms and Gated Recurrent Units (GRU) to learn the differences between normal and abnormal traffic. They used Convolutional Neural Networks (CNN) to capture local features and fully connected layers to classify traffic [13]. Lan Li and Gyungho Lee utilized wavelet analysis to monitor changes in energy distribution over time to identify attacks [14]. Neha Agrawal and Shashikala Tapaswi combined temporal domain analysis of Traffic Flow Behavior Graphs with frequency domain analysis of power spectral density to detect low-rate DDoS attacks [15].

Although these methods eliminate the dependency on IP information, they do not segregate services that are subjected to high-rate and low-rate attacks based on ports. Instead, they process all traffic uniformly, which is detrimental to latency reduction and results in poor real-time performance.

To address the issues present in these existing methods, we have designed the PAHDD system. Without relying on IP information, the PAHDD system adaptively locks high-risk attack ports, isolates services experiencing high-rate DDoS attacks, services under low-rate DDoS attacks, and well-functioning services based on ports, and conducts tiered detection of high-rate and low-rate DDoS attacks.

III. DESIGN OF PAHDD

In network environments characterized by NAT and stringent confidentiality measures, destination IP information is often unavailable or unreliable. This limitation significantly reduces the applicability and effectiveness of existing DDoS detection methods. Therefore, our primary design objective is to develop a system that can effectively detect DDoS attacks without relying on IP information. In the design section of this paper, we provide a detailed exposition of the port-based adaptive high-speed traffic DDoS real-time detection (PAHDD) System.

A. Overview

NAT technology enhances the privacy of network devices, rendering destination IP information inaccessible, as illustrated in Fig. 2. This renders most IP-based DDoS detection methods ineffective [2]. The PAHDD system, however, can lock high-risk ports using Adaptive Port-based Risk Locking, even when the mapped ports are unknown. This allows for effective DDoS attack detection without relying on IP information.

B. Overview

As illustrated in Fig. 3, the PAHDD system's main framework comprises six modules. When network packets enter the system, they first pass through the Buffer and RFC Detection modules, where they are processed into flow units and subjected to an initial screening. The screened traffic is then directed to the core module, Adaptive Port-based Risk Locking, where it is categorized into different branches based

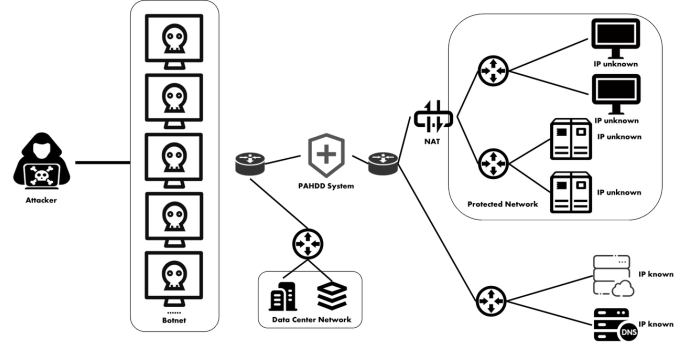


Fig. 2. Schematic Diagram of Detection Scene.

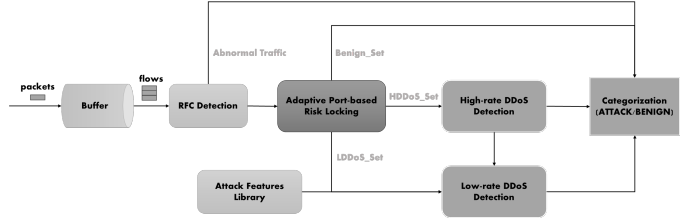


Fig. 3. PAHDD System Main Architecture Diagram.

on risk and port for targeted processing. Within these branches, the HDDoS Detection and LDDoS Detection modules utilize traditional methods and machine learning techniques to analyze high-speed and low-speed traffic, respectively.

C. Architecture

Buffer. The Buffer module is equipped with queue buffers that restore packets into flow units based on their flow IDs. This design allows the system to conduct real-time detection while minimizing latency.

RFC Detection. In this module, we swiftly process network packets that fail to meet protocol standards or contain clear vulnerabilities. Malformed packets—resulting from missing necessary fields, incorrect field values, or other deviations from protocol standards—are identified as malicious traffic. Such packets can be detected by analyzing fields like *TCP_Flags*.

Adaptive Port-based Risk Locking. In this module, source and destination ports are categorized into *Src_HDS*, *Src_LDS*, *Src_BS*, *Dst_HDS*, *Dst_LDS*, and *Dst_BS*. For each source and destination port, a *Flow_State_Vector* class is established. This class records the flow state vector and other relevant information for the traffic passing through the port, with a time window size of T .

1) *Initialisation Phase*: Public service ports 53 and 80 are particularly vulnerable to flooding attacks, so they are initially placed in the *HDDoS_Set*.

2) *Adaptive Risk Port Locking Phase*: High-rate DDoS attacks are distinguished by their substantial traffic volume, enabling adaptive locking of high-rate attack ports based on traffic thresholds. The traffic threshold is determined based on normal traffic conditions. In the implementation of PAHDD,

we select the *maximumvalue + variance* of the number of packets within the time window T in the background traffic as the traffic threshold. Service ports exceeding this threshold are deemed to contain high-speed DDoS traffic and are added to the *HDDoS_Set*. The traffic from these ports is then redirected to the HDDoS Detection module for rapid detection.

Benign_Set represents a high level of trust, akin to authorization. During the algorithm's initialization phase, granting unrestricted access to any traffic must be done with caution. The average traffic state vector of the background traffic, *Benign_mean_FSV*, is used as the clustering center. If the *FSV* value deviates from the threshold, it is flagged as a potential DDoS attack. Adhering to the principle of caution in authorization, *Src_BS* and *Dst_BS* are initialized as empty sets. *BS* uses the traffic state vector $\langle pps, bps, port_entropy \rangle$ to monitor and update. Unlike the *packet_count* required for *HDS* locking, the traffic state vector calculation only counts the source and destination ports of a flow once to effectively distinguish between attacked and non-attacked port services.

Algorithm 1 provides a detailed description of this module, with the symbols defined in Table I. If a flow's source port and destination port are neither in *HDS* nor *BS*, they are classified as *LDS*.

TABLE I
GLOSSARY OF SYMBOLS.

pps	packet per second
bps	bits per second
port_entropy	The distribution entropy of another port
BS	Benign Set
HDS	High-rate DDoS Set
LDS	Low-rate DDoS Set
FSV	Flow State Vector

HDDoS Detection. The HDDoS Detection module maintains a sliding average of key traffic features for each port within *HDS*. When a new flow enters this module, if its feature values deviate from the sliding average of the critical traffic features, the flow is considered not to represent the high-speed attack traffic currently affecting that port. We employed the XGBoost algorithm [19] to analyze feature importance within the training set, as illustrated in Fig. 4. The analysis revealed that *ip_id* and *src_port* are highly significant due to the substantial presence of high-rate DDoS attack traffic in the dataset, characterized by highly random source port numbers and *ip_ids*. High-rate attacks flood the victim's network devices with vast amounts of malicious traffic. Therefore, in Algorithm 2, we select *ip_id* and *another_port* as critical flow features. Given that the characteristics of low-speed attack traffic differ from those of high-rate attacks, traffic identified by the HDDoS Detection module may indicate the presence of low-rate attacks. As a result, LDDoS Detection should be conducted immediately following HDDoS Detection. The detailed process of this module is presented in Algorithm 2.

Algorithm 1 APRL

Class *Flow_State_Vector*:

State: $\langle pps, bps, port_entropy \rangle$
Timestamp_ptr, *Packet_count*
HDDoS_thresh, *Benign_mean_FSV*
Benign_FSV_distance_thresh

Initial Stage:
Initialize *src_HDS*, *src_LDS*, *src_BS*
Initialize *dst_HDS*, *dst_LDS*, *dst_BS*
Initialize *FSV_dict*
 $T = 200ms$

Running Stage:

- 1: **while** not end **do**
- 2: $Flow = \text{buffer.dequeue}()$
- 3: Update *FSV_dict*
- 4: **if** $Flow.timestamp > FSV_dict[Flow].Timestamp_ptr + T$
then
- 5: Update *src_HDS*, *src_LDS*, *src_BS*
- 6: Update *dst_HDS*, *dst_LDS*, *dst_BS*
- 7: $FSV_dict[Flow].Timestamp_ptr = Flow.timestamp$
- 8: **end if**
- 9: **if** port in *HDS* **then**
- 10: *HDDoS_Detection()*
- 11: *LDDoS_Detection()*
- 12: **else if** port in *LDS* **then**
- 13: *LDDoS_Detection()*
- 14: **end if**
- 15: **end while**

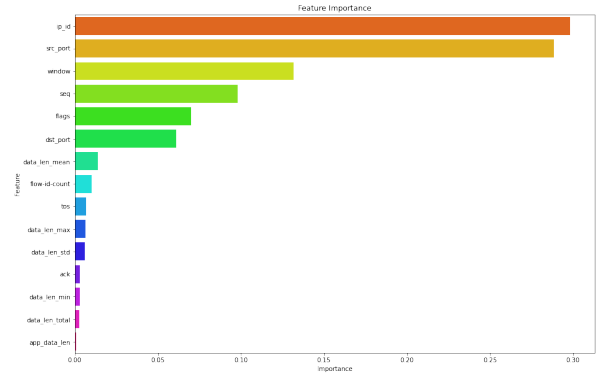


Fig. 4. Importance Map of Input Model Features.

LDDoS Detection. Low-speed DDoS traffic resembles benign background traffic and often incorporates multiple attack types in real-world scenarios, making it difficult for traditional methods to achieve high accuracy. In contrast, machine learning techniques can integrate data from various dimensions, allowing for a more nuanced identification of attack characteristics that traditional methods may overlook. This capability can significantly reduce the false positive rate and improve overall detection accuracy.

Algorithm 2 HDDoS Detection

```

1: if dport in Dst_HDS then
2:   if sport < HD_port_thresh and
     ip_id < HD_ip_id_thresh then
3:     return HDDoS
4:   end if
5:   update HD_port_thresh, HD_ip_id_thresh
6: else if sport in Src_HDS then
7:   if dport < HD_port_thresh and
     ip_id < HD_ip_id_thresh then
8:     return HDDoS
9:   end if
10:  update HD_port_thresh, HD_ip_id_thresh
11: end if
12: return Not HDDoS

```

After evaluating the performance of various machine learning algorithms, we chose the Random Forest algorithm for its superior results. Random Forest employs the Bagging method, constructing multiple decision trees and aggregating their votes to make predictions. This approach effectively mitigates the overfitting risk inherent in single models and is well-suited for multi-dimensional traffic features. Furthermore, in the train.pcap dataset, the proportion of attack traffic significantly surpasses that of normal traffic, and Random Forest’s resampling technique is particularly effective in addressing such imbalanced data.

In the LDDoS Detection module, as outlined in Algorithm 3, we first match the traffic against the attack feature database before employing the Random Forest algorithm for prediction. This preliminary step is designed to reduce the computational cost of the machine learning model and improve the overall accuracy of the assessments.

Algorithm 3 LDDoS Detection

```

1: if Match_attack_feature then
2:   return LDDoS
3: end if
4: calculate feature_vector
5: return model.predict(feature_vector)

```

Attack Features Library. The attack feature database is constructed from the expertise of network security engineers, enabling the swift identification of previously encountered attacks and reducing the overall system load. As the system gains more experience with various attacks, its robustness improves. In the implementation of PAHDD, we integrate features derived from this accumulated knowledge into the LDDoS Detection module, resulting in improved detection effectiveness.

IV. EXPERIMENTAL RESULTS

A. Evaluation Metrics

In line with the conventions outlined in references [17], [18], we will assess the constructed high-speed traffic detection sys-

tem using the *F1-score*, a metric that evaluates the accuracy of binary classification models. The subsequent section will introduce the key evaluation parameters.

Precision: The ratio of correctly predicted positive observations to the total number of predicted positives, measuring the system’s ability to avoid false positives.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

Recall: The ratio of correctly predicted positive observations to the total number of actual positives in the dataset. This metric assesses the system’s sensitivity.

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

F1-score: The harmonic mean of precision and recall, used to assess the overall performance of the system comprehensively.

$$F1-score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (3)$$

where $TP = TruePositive$, $FP = FalsePositive$, $TN = TrueNegative$, $FN = FalseNegative$.

B. Experimental Configuration and Dataset

Our system was tested in the following environment: *Ubuntu 22.04*, *Python 3.12.3*, without *GPU*. To rigorously validate the effectiveness of our system, we primarily utilized the Aitrans2024 competition dataset [16], which consists of 99% mixed attack traffic and 1% benign background traffic. The source and destination IP addresses in this dataset have been randomly perturbed to approximate a high-privacy scenario where destination IP information is unavailable. Table II presents the traffic statistics for the training and testing sets.

TABLE II
TRAFFIC STATISTICS FOR THE TRAINING AND TEST SETS.

		[80, 53]	else	Percentage
Train	Attack	904966	318585	98.92%
	Benign	1528	11806	1.08%
Test	Attack	3741412	530540	99.40%
	Benign	2422	23538	0.60%

C. Results

We trained a Random Forest algorithm model for LD-DoS detection using the training dataset train.pcap from the Aitrans2024 competition. The experimental results show that our system achieved an accuracy of 99.87%, a precision for positive predictions of 93.93%, and an *F1-score* of 89.01%. The training dataset contains approximately 1.23 million records, while the test dataset comprises about 4.5 million records, demonstrating the excellent generalization performance of the PAHDD system.

We separately tested the average latency of the HDDoS Detection and LDDoS Detection modules, which were 0.84 microseconds and 915.52 microseconds, respectively. The latter is 1090 times longer than the former due to the extended

computation time required for inference by the machine learning model in the LDDoS Detection module. The Adaptive Port-based Risk Locking module locks high-rate attacked ports, allowing a significant portion of high-rate attacks to be processed first by the HDDoS Detection module, substantially reducing latency. The actual average inference latency in operation, including delays introduced by the Buffer module and the APRL module, is 284.61 microseconds.

D. Discussion

The Buffer module of the PAHDD system is designed to achieve effective detection of low-rate DDoS attacks by processing data flow as a unit for the subsequent system, but this results in a multiplicative effect on detection latency, with an average packet detection latency of 284.61 microseconds.

It is important to clarify that the PAHDD system is a DDoS detection system, capable of taking preventive measures against detected malicious traffic based on the detection results, issuing early warning information to network security engineers, or directly cleansing the traffic.

Besides the traditional NAT technology targeted in this paper, there are also Twice NAT and NAPT (Network Address Port Translation) technologies. In addition to completing the PAHDD system for high-privacy networks under traditional NAT technology, this paper also proposes research directions for networks under these two other NAT technologies. By improving DDoS detection in these contexts, a more mature DDoS detection system can be formed for NAT environments [2].

V. CONCLUSION

In this paper, we present a port-based adaptive high-speed traffic DDoS real-time detection system, PAHDD, designed for high-privacy networks where precise IP addresses cannot be obtained. This system innovatively conducts DDoS detection based on ports, utilizing an adaptive risk port locking algorithm to categorize ports into high-rate DDoS ports, low-rate DDoS ports, and benign ports. This enables classified processing and real-time detection of traffic. The system does not rely on IP information, thus meeting the DDoS detection needs of high-privacy network environments. Experimental results show that the PAHDD system achieves an accuracy of 99.87% on the Aitrans2024 competition dataset [16].

The system can be applied in scenarios where accurate IP addresses cannot be obtained, demonstrating excellent performance in detecting DDoS traffic in high-privacy networks. By performing detection based on ports rather than relying on the IP information of the protected network segment, PAHDD provides a new research approach for DDoS detection technology in high-privacy network environments.

REFERENCES

[1] Haseeb-ur-rehman, R.M.A.; Aman, A.H.M.; Hasan, M.K.; Ariffin, K.A.Z.; Namoun, A.; Tufail, A.; Kim, K.-H. High-Speed Network DDoS Attack Detection: A Survey. *Sensors* 2023, 23, 6850.

[2] Sourour M, Adel B, Tarek A. Network security alerts management architecture for signature-based intrusions detection systems within a NAT environment[J]. *Journal of Network and Systems Management*, 2011, 19(4): 472-495.

[3] Alibaba Cloud, "Anti-DDoS Origin Basic," Alibaba Cloud, <https://www.alibabacloud.com/help/zh/nat-gateway/security-and-compliance/anti-ddos-origin-basic> (accessed July, 2024).

[4] Adedeji, K.B.; Abu-Mahfouz, A.M.; Kurien, A.M. DDoS Attack and Detection Methods in Internet-Enabled Networks: Concept, Research Perspectives, and Challenges. *J. Sens. Actuator Netw.* 2023, 12, 51.

[5] Villalobos, J. J. , I. Rodero , and M. Parashar . "An Unsupervised Approach for Online Detection and Mitigation of High-Rate DDoS Attacks Based on an In-Memory Distributed Graph Using Streaming Data and Analytics." *ACM* (2017):103-112.

[6] David J , Thomas C .Efficient DDoS Flood Attack Detection using Dynamic Thresholding on Flow-Based Network Traffic[J].*Computers & Security*, 2019, 82(MAY):284-295.Doi:10.1016/j.cose.2019.01.002.

[7] Li, Ke , et al. "Distinguishing DDoS Attacks from Flash Crowds Using Probability Metrics." 2009 Third International Conference on Network and System Security IEEE, 2009.

[8] Bhuyan, Monowar H. , D. K. Bhattacharyya , and J. K. Kalita . "Information metrics for low-rate DDoS attack detection: A comparative evaluation." 2014 Seventh International Conference on Contemporary Computing (IC3) 0.

[9] Sachdeva, Monika , K. Kumar , and G. Singh . "A comprehensive approach to discriminate DDoS attacks from flash events." *Information Security Technical Report* 26.FEB.(2016):8-22.

[10] Toklu S , Simsek M .Two-Layer Approach for Mixed High-Rate and Low-Rate Distributed Denial of Service (DDoS) Attack Detection and Filtering[J].*Arabian Journal for Science & Engineering*, 2018, 43(12):7923-7931.

[11] Liu X, Ren J, He H, et al.Low-rate DDoS attacks detection method using data compression and behavior divergence measurement[J].*Computers & Security*, 2021, 100:102107.Doi:10.1016/j.cose.2020.102107.

[12] Zhou, Y., Hu, G., & He, W. (2009, July). Using graph to detect network traffic anomaly. In 2009 International Conference on Communications, Circuits and Systems (pp. 341-345). IEEE.

[13] X. Yuan, C. Li and X. Li, "DeepDefense: Identifying DDoS Attack via Deep Learning," 2017 IEEE International Conference on Smart Computing (SMARTCOMP), Hong Kong, China, 2017, pp. 1-8, doi: 10.1109/SMARTCOMP.2017.7946998.

[14] Li, et al. "DDoS attack detection and wavelets." *Computer Communications and Networks*, 2003. ICCCN 2003. Proceedings. The 12th International Conference on 2003.

[15] Agrawal, N., & Tapaswi, S. (2020). Detection of low-rate cloud DDoS attacks in frequency domain using fast hartley transform. *Wireless Personal Communications*, 112(3), 1735-1762.

[16] "Aitrans2024 Competition Dataset," 2024.[Online]. Available: <https://final.thu.aitrans2024.online/>. [Accessed: June, 2024].

[17] S. H. Park, J. M. Goo, and C. H. Jo, "Receiver operating characteristic (ROC) curve: practical review for radiologists," *Korean Journal of Radiology*, vol. 5, no. 1, p. 11, 2004.

[18] M. W. Powers, "Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation," *Journal of Machine Learning Technologies*, vol. 2, no. 1, pp. 37-63, 2011.

[19] Chen T, Guestrin C. Xgboost: A scalable tree boosting system[C]//Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining. 2016: 785-794.